

Data Cleaning Documentation

Introduction

This document outlines the data pre-processing steps applied to prepare the dataset for analysis. It covers data cleaning, handling of missing values, formatting, and transformation processes to ensure consistency, accuracy, and readiness for downstream analysis.

Key Columns and Their Meanings

Column Name	Meaning
Receipt Number	Unique identifier for each import transaction.
Importer	Name or ID of the importing company or individual.
HS Code	Harmonized System Code for classifying traded products.
Country of Origin	Country where the imported goods were manufactured or produced.
Country of Supply	Country from which the goods were shipped.
FOB Value (N)	Free On-Board value representing the cost of goods excluding shipping and insurance.
CIF Value (N)	Cost, Insurance, and Freight value representing the total import value.
Tax Amount (N)	Total amount of tax paid for each import transaction.
Container Number	Unique identifier for the shipping container used.
Container Size	Size of the shipping container, indicating volume capacity.
Weight (kg)	Mass of the imported goods in kilograms.
Receipt Date	Date the goods were received or cleared by customs.
Customs Office	Name or ID of the customs office handling the transaction.
Transaction Type	Indicates the type of trade, e.g., import or export.

Column Name	Meaning
High-Risk Country	Boolean flag indicating if the import is from a high-risk country based on compliance criteria.

Data Cleaning Steps and Techniques

The dataset was cleaned using Excel, Power Query techniques and Python to ensure data consistency and accuracy. The process involved:

Handling missing values and invalid entries:

- An invalid record in the number of container column was replaced with the median due to the skewness of the data.
- Receipt date was replaced with the modal date since the missing dates only account for about 1.6% of our data which would not significantly impact our analysis.
- Replaced missing values with “unknown” in the following columns; receipt number, container number, and container size.

Converting data types for accurate analysis

- Receipt date was converted to datetime data type.
- Importer and HS Code were converted to strings since they are unique identifiers.

Handling outliers

Winsorization method was used in FOB values, CIF values, total tax and mass (kg) to cap the top 17% of the data to reduce the effect of outliers.